Aditya Venkata Subramanian (Aditya V)

aditya22@iitk.ac.in +91-9380-127-260

Aditya V Aditya-Venkat aditya-venkat.github.io

Education

BS Indian Institute of Technology Kanpur, Statistics and Data Science — 2022 – 2026
CPI: 9.4/10

- **Awards:** Academic Excellence Award (2022, 2023 & 2024)
- **Honors Track Student**
- **Minor:** Machine Learning and Applications

Grade XII National Centre for Excellence, Bengaluru — **Score:** 97.2% | 2022

Preprints

Links to below papers will be updated on my personal website as they become available

Benchmarking of Spatial Transcriptomics Methodologies
Ajita Shree, Aditya V, Tanush Kumar, Hamim Zafar (will be available soon on biorxiv)
Comprehensive benchmarking of 20+ spatial transcriptomics methods across simulated datasets, introducing novel metrics and simulation techniques. Poster accepted at RECOMB 2025, Seoul

A Bayesian Bivariate Load-sharing Model
Aditya V, Debasis Kundu (Manuscript under preparation)
Proposed a Bayesian extension of classical load-sharing models with multiple baseline distributions and MCMC-based credible interval estimation for reliability parameters

Research Experience

Benchmarking of Spatial Transcriptomics Methodologies | Prof. Hamim Zafar | IIT Kanpur (*Sep '23 - Nov '25*)
Second-authored a poster accepted to RECOMB 2025, Seoul

- Built pipelines of **10+** different Spatial Deconvolution methodologies and **10+** Spatial Domain detection methods
- Evaluated the methods on **35+** simulated datasets from **4+** different categories on over **80 Gigabytes** of data
- Designed a **novel spatial bivariate statistic** and implemented **simulation methods** for **imaging-based data**
- Developed essential, consistent insights of method performance using **10+** novel, standard and spatial metrics

Bayesian Bivariate Load-sharing Model | Prof. Debasis Kundu | IIT Kanpur | (*Jan '25 - present*)

- Implemented a **bivariate load-sharing model**, overcoming flaws in traditional models and tested estimation approaches on simulated datasets achieving **>95% coverage** rates.
- Created novel **Bayesian extensions** for exponential, weibull and piecewise constant hazard baselines using suitable priors, and demonstrated usability
- Implemented **MCMC sampling techniques** to estimate credible intervals of parameters of interest overcoming **flaws of MLE-based estimators**

ML and Cancer Research Internship| Dr. Karthik | TenSixty Bio | Massachusetts (remote) (*Jun '24 - Aug '24*)

- Worked with the co-founders to build the **spatial binding** component of their cancer therapeutics pipeline
- Built an **unsupervised Computer Vision algorithm** to quantify IHC staining at cellular resolutions
- Used **clustering-based approaches**, leveraging **cell-segmentation models** to measure antibody expression
- Set up a process hosted on **Amazon Web Services** to extract and analyze antibody expression across samples

Selected Projects

Sampling-based Bayesian Inference Unlearning | MTH707 | Prof. Dootika Vats | IIT Kanpur | (*Jan '25 - Apr '25*)

- Investigated **MCMC-based machine unlearning** algorithms for **Bayesian inference models** without full retraining
- Explored **KL-divergence** based inference unlearning and potential for a novel extension to **JSD-based unlearning**

- Designed methods using **low-rank Hessian approximations** to improve unlearning accuracy by integrating **Stochastic Gradient Riemann Langevin Dynamics** with KL-divergence based unlearning
- Conducted **simulation studies** validating posterior recovery, and error bounds under data removal scenarios

Multimodal Cell Type Imputation | Prof. Hamim Zafar | IIT Kanpur | (*Sep '25 – present*)

- Built **spatial joint embeddings** using GAN-based models conditioned with **niche-based inference parameters**
- Applied **non-parametric classifiers** (Spearman correlation, multi-class p-tests) for cell-type inference
- Developing a **graph-based dual-autoencoder with adversarial training** for improved spatial representations.

Technical Skills

Languages: Python, C, C++, R, SQL **Other Tools:** Git, Markdown, AWS, NextFlow, RShiny, LaTeX
Libraries: Numpy, Pandas, Scikit-Learn, Matplotlib, Seaborn, HuggingFace, PyTorch, OpenCV, XML.eTree

Relevant Coursework

Legend: ⋆⋆ Outstanding Performance ⋆ Excellent Performance † Ongoing Course

Foundational courses	Methodology & Modeling	Computational Data Science
Introduction to Probability Theory	Linear Regression and ANOVA ⋆	Introduction to Machine Learning ⋆⋆
Matrix Algebra & Linear Estimation	Non-Linear Regression ⋆⋆	Data Science Labs ⋆
Theory of Statistics	Time Series Analysis ⋆	Statistical Computing
Stochastic Processes ⋆	Non-Parametric Inference ⋆	Data Structures and Algorithms
Single & Several Variable Calculus ⋆	Introduction to Bayesian Analysis	Fundamentals of Computing ⋆
Ordinary Differential Equations ⋆	Markov Chain Monte Carlo ⋆	Computational Genomics ⋆
Probability Theory †	Probabilistic Machine Learning †	Differential Privacy in ML †

Scholastic Achievements

- Secured **All India Rank 817** in JEE Advanced & **All India Rank 721** in JEE Mains among 1 Million candidates *2022*
- Secured **All India Rank 3** in IISER Aptitude test amongst 20 thousand candidates from across the country *2022*
- Secured **All India Rank 641** in KVPY SA amongst 50 thousand candidates and became eligible for scholarship *2021*

Mentorship

Sound Refine | Project Mentor | BCS, IIT Kanpur | (*May '24 - July '24*)

- Engineered a pipeline for **noise reduction**, **transcription**, and **summarization** on data built via audio mixing
- Led mentees through the basics of Python, Machine Learning, Audio processing, **CNNs**, **U-Nets**, and **Transformers**

Academic Mentor, PHY115 | Counselling service, IIT Kanpur (*Apr '23 - Apr '24*)

- Guided **600+** freshers through a course on oscillation and waves and was a personal mentor to **16** students
- Delivered a lecture to a classroom of **300** students on the wave equation and Fourier transforms for their end-term

Work Experience

Associate Consulting Intern | Bain & Co. | Mumbai (Onsite) (*May '25- Jul '25*)

- Designed **automated models** to recommend organizational restructuring actions across **40+ client accounts**
- Developed **quantitative techniques** to estimate value creation potential ("size of prize") under limited data
- Optimized account-level overheads by **clustering 40+ accounts**, identifying cost drivers, and benchmarking

Positions of Responsibility

Coordinator |Debating Society, IITK (*Apr '24 - Apr '25*)

- Initiated and convened the first and second editions of IITKPD with **80+** participants, and a budget of **INR 2,50,000**
- Organised **UPDL'24** at IIT Kanpur, a national **3-tier** tournament with **3000+** participants from over **8** states
- Improved debate training quality by introducing external debate coaches and brought in funds over **INR 1,00,000**